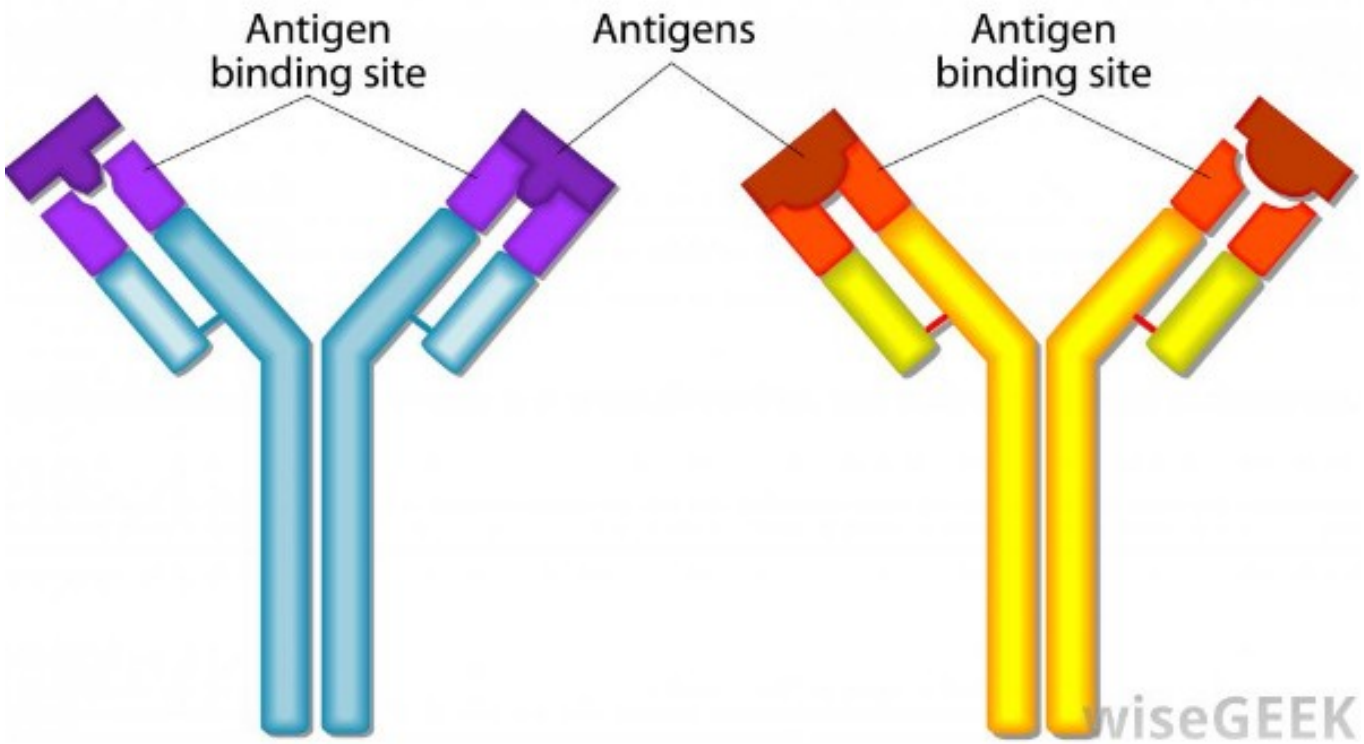
Specific Resistance to Infection



Specific Immune Response and Types of Immunity

Specific Defences against Disease

- Tailored to each individual pathogen to overcome infection AND to respond quickly on subsequent encounters to clear infection before symptoms develop.

- Creates “memory” to prevent getting it again.
  - Eg: Once you’ve had measles, you’re immune. You don’t get it a second time (usually)

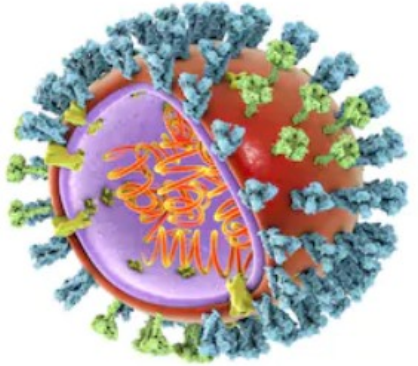
- Involves Macrophages and other phagocytes
  - Also involved in non-specific immunity
  - Engulf pathogens and present antigen on surface to lymphocytes
  - Are therefore “Antigen Presenting Cells”.

- Involves Lymphocytes
  - Also involved in non-specific immunity
  - Receive antigen and trigger specific immunity (cell-mediated and antibody mediated)

Learning Aim: Define “specific defences against disease” and list the main cell types involved

Specific Defences – What are Antigens?

- Definition: A substance capable of producing an immune response:
  - The part of the pathogen that triggers the immune response: “antigenic site”.
  - May not be a pathogen: Allergy occurs when the immune system reacts to a substance that is not harmful. Vaccines contain antigen to stimulate immunity without disease.
  - Usually large molecules
  - Could be a variety of things:
    - Whole microorganism
    - Part of the protein coat
    - A toxin produced by a pathogen
    - Foreign tissue (eg donor blood or tissue)

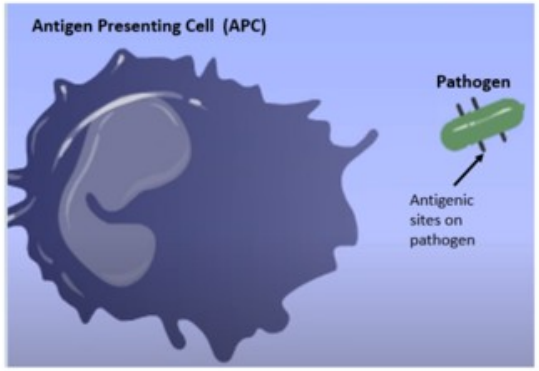


  - Immune system can distinguish between self-antigens and non-self antigens

Learning Aim: Define “antigen” and discuss what parts of pathogens can be antigenic

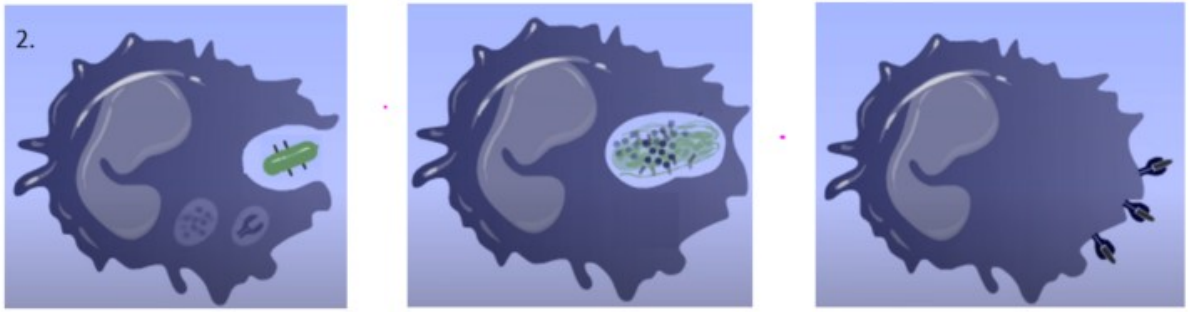
Specific Defences – Antigen Presenting Cells (APC)

- These cells are phagocytes, that ingest pathogens as part of the non-specific response.
- they then present the antigen (part of the pathogen that the immune system recognises) on their surface.
- They "present" the antigen to lymphocytes which are involved in the specific immune response.
- Some examples of APC include:
  - Macrophages
  - Some lymphocytes
    (have several roles, not just APC)
  - Dendritic cells (not a focus in Y12 HB)
- Note: many of one type of pathogen enter the body at once, so while the APCs are doing their work, others of the specific pathogen are starting to cause disease.



1. Invading Pathogen encounters APC in extracellular fluid surrounding tissues or in lymph node.

- Pathogen eg: virus, bacterium
- APC eg: macrophage, lymphocyte



The APC engulfs the pathogen ...

... and processes it...

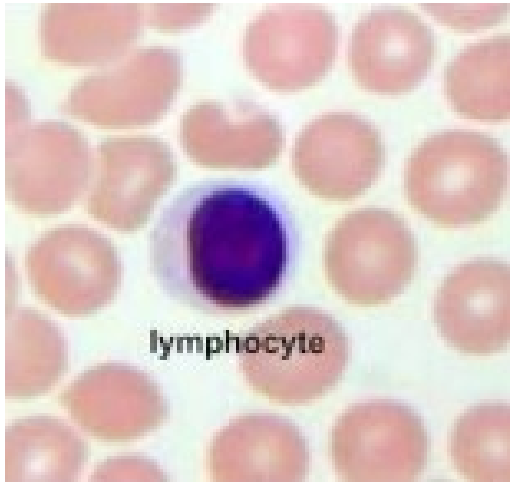
... and presents the antigen on its surface

Learning Aim: Describe the function of Antigen Presenting Cells

Specific Defences - Lymphocytes

- Lymphocytes: white blood cells that are involved both non-specific and specific defence
- Can be found:
  - In blood circulation
  - In tissues
  - In lymph nodes

- Produced in bone marrow.



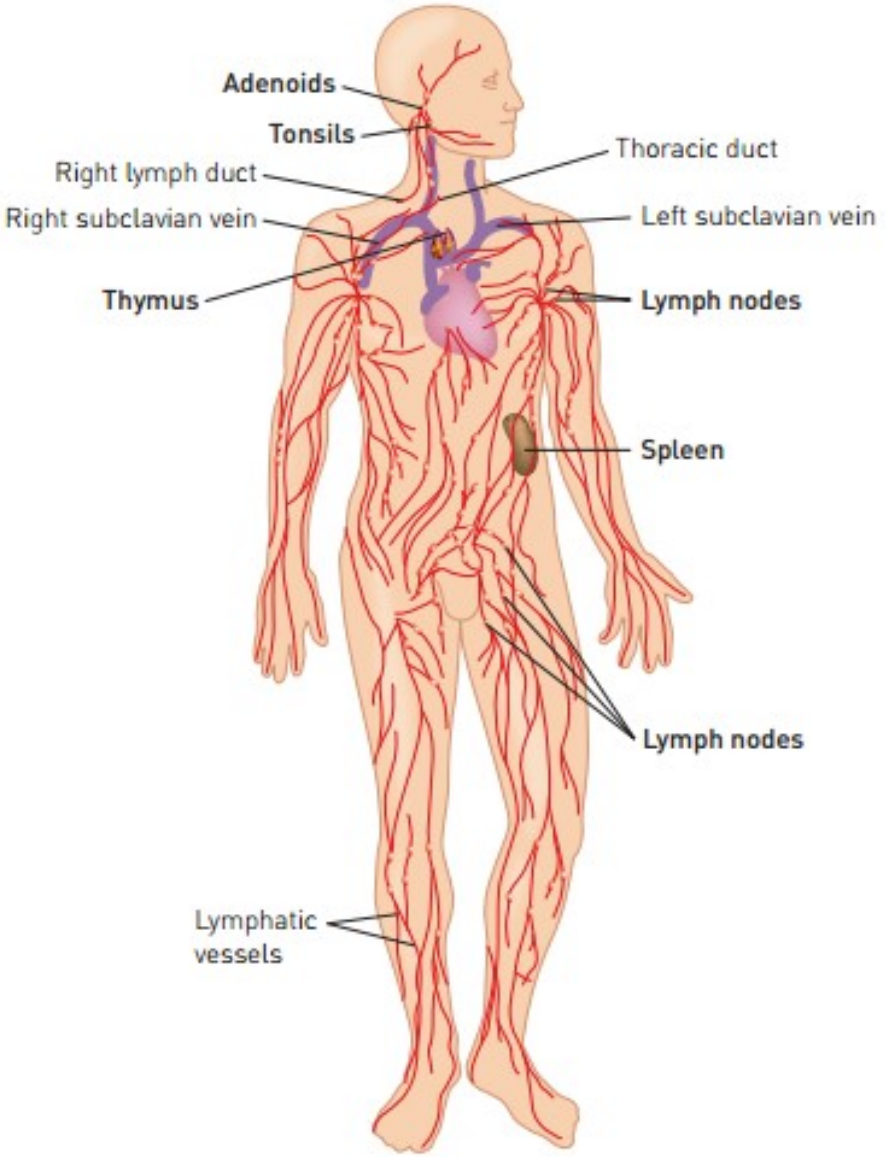
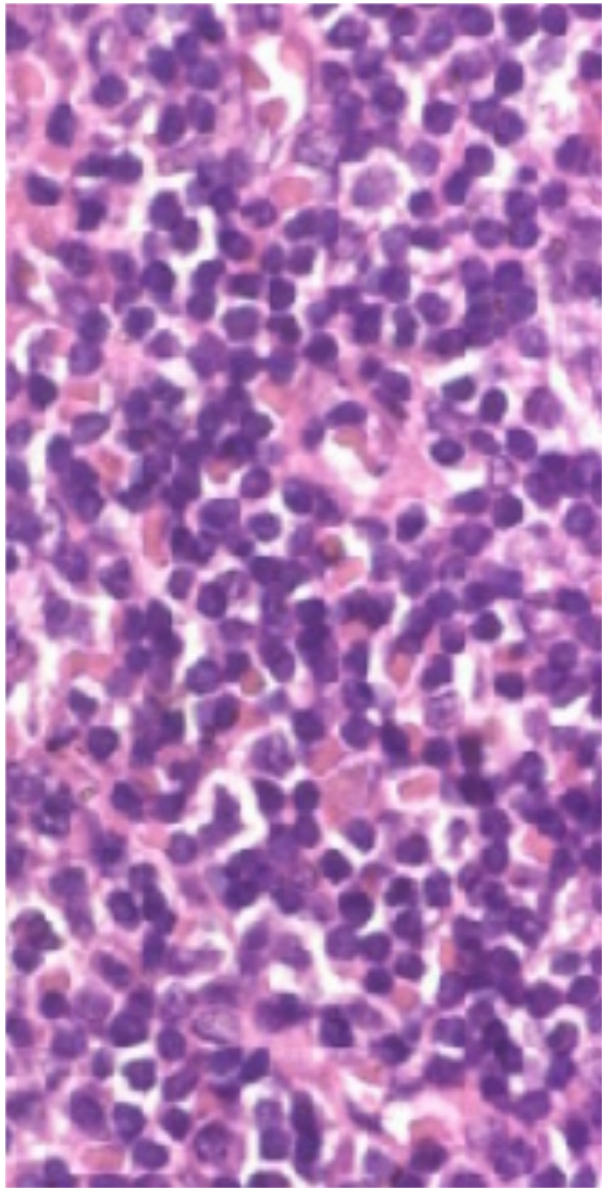


Figure 11.1 Lymphoid tissue occurs throughout the body.

Learning Aim: Describe where lymphocytes are produced and where they can be found

Specific Defences - Lymphocytes

- 2 types involved in specific immunity:
  - B- Lymphocytes (B-cells) – antibody mediated (humoral) immunity
  - T- Lymphocytes (T-cells) – cell mediated immunity
  - Both produced by bone marrow, but mature in different ways
- B-cells:
  - Made in bone marrow
  - **Mature in bone marrow**
  - Move to lymphoid tissues and blood
  - Involved in *anti-body mediated (humoral) immunity*
- T-Cells
  - Made in bone marrow
  - **Mature in Thymus**
  - Move to lymphoid tissues and blood
  - Involved in *cell-mediated immunity*
- B and T cells make up the majority of the lymphoid tissues



Lymphocytes in lymphoid tissue

Learning Aim: Compare B- and T- lymphocyte maturation and function.

Specific Immunity

- APC and lymphocytes work together to identify and neutralise specific pathogens, and form memory cells so that long-term immunity to those pathogens is achieved.

- Two types of specific immunity:
  - Antibody-Mediated (Humoral) – deals with pathogens before entry to cells
  - Cell-Mediated Immunity – deals with pathogens that have entered cells

- Effective for pathogens that produce consistent antigenic sites.
  - EG: Measles virus has consistent antigenic sites.
    - The first time measles is encountered, APC present the antigen to lymphocytes which act specifically to destroy the measles virus.
    - Memory lymphocytes are produced which remember the antigenic site for the measles virus so it can be destroyed before it causes disease the next time it is encountered.

  - EG: this will not work as well for the virus that causes the common cold.
    - the antigenic site on the cold virus varies, so cannot be recognised by memory cells in subsequent encounters

Specific Immunity: Antibody-Mediated (Humoral) Immunity

- Also known as B-Cell Immunity
- Involves B-lymphocytes
  - Made in bone marrow
  - Mature in bone marrow
  - Have thousands of unique receptors on surface
- Deals with pathogens that have not yet entered body cells.
  - Eg viruses before entry to cell
  - Bacteria
  - Some toxins
- Production and release of **antibodies** into blood and lymph

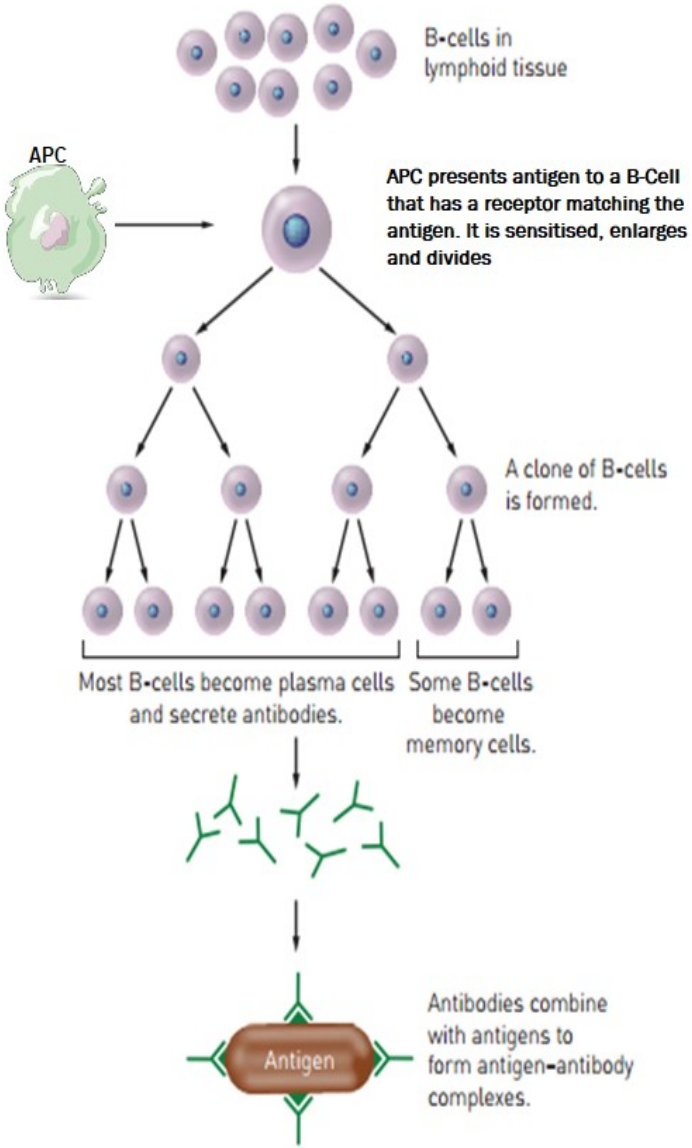


Figure 11.5 Events in the antibody-mediated immune response

Learning Aim: Describe in detail the function and processes involved in antibody-mediated (humoral) immunity

Specific Immunity: Antibody-Mediated (Humoral) Immunity

1: B-lymphocytes in lymphoid tissue

2: Antigen Presenting Cell (APC) which has engulfed and destroyed pathogen, presents the antigenic site for that pathogen on its surface.

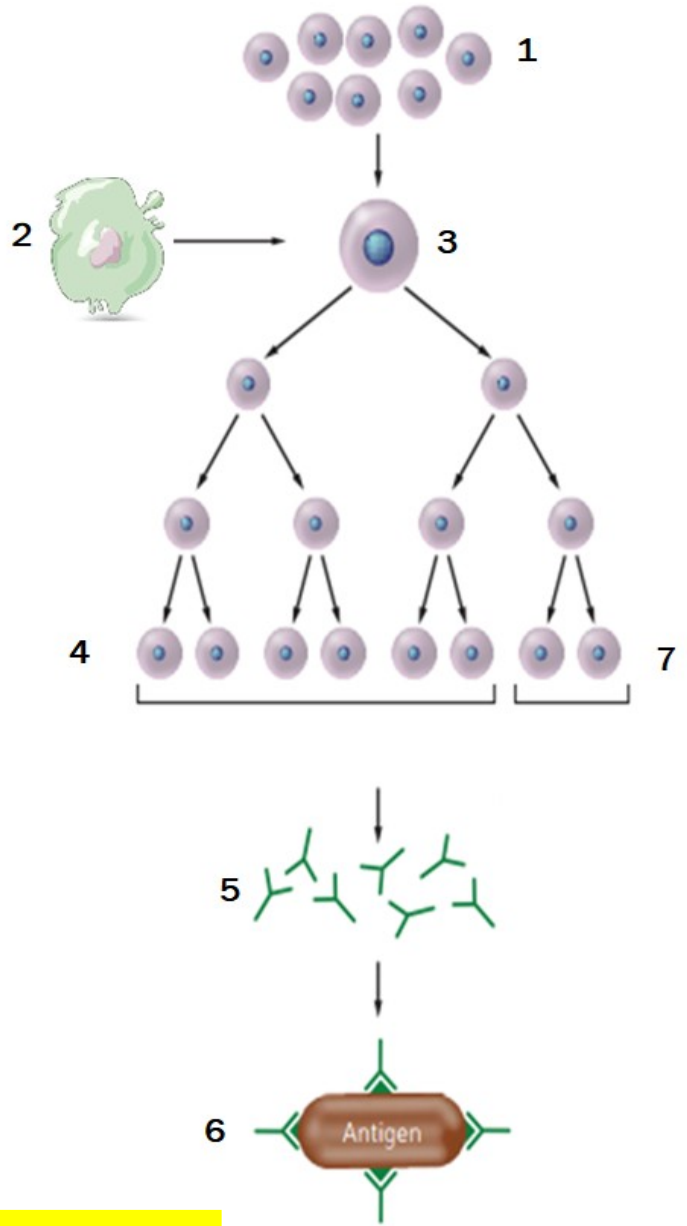
3: The APC presents the antigen to a B-cell with a matching receptor. The B-cell becomes sensitised, enlarges and divides, producing many more sensitised B-cells.

4: Most of these sensitised B-cells become plasma cells.

5: The plasma cells produce antibody specific for the antigen originally presented by the APC.

6: The antibody is released, and is then able to bind to antigenic sites on the same pathogens as the one originally presented by the APC, neutralising the pathogen.

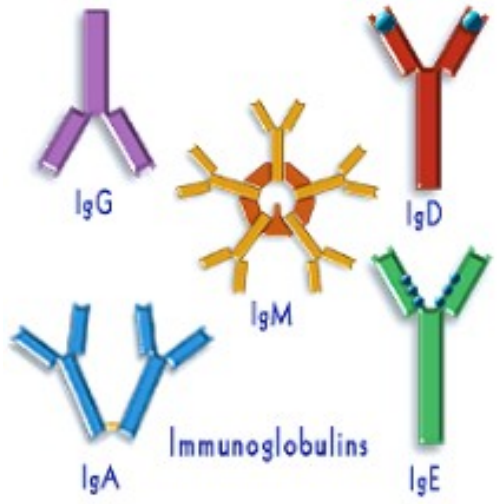
7: Sensitised Memory B-cells remain in circulation, so that the response proceeds more quickly if the pathogen is encountered again.

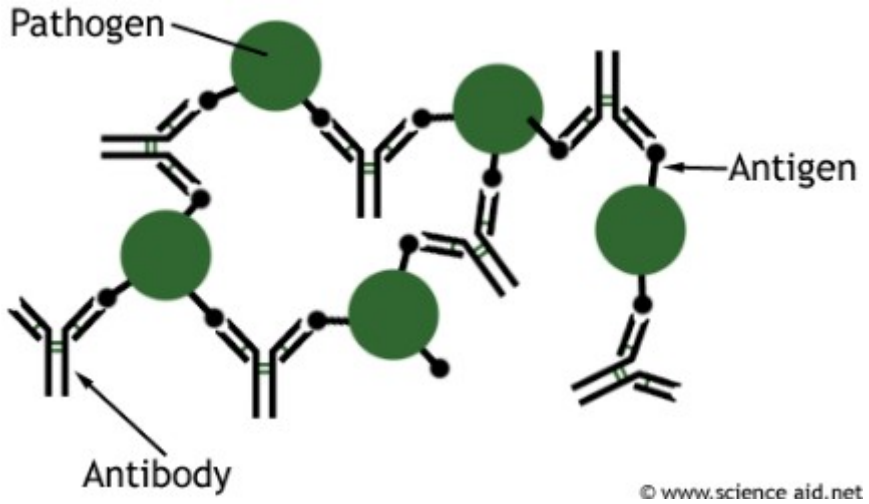


Learning Aim: Describe in detail the function and processes involved in antibody-mediated (humoral) immunity

Specific Immunity: Antibodies

- A special type of protein with active sites/receptors that bind to specific antigen. Forms antigen-antibody complex.
- Produced during Antibody-mediated immunity in response to non-self antigens.
- Antibodies are also called *immunoglobulins (Ig)*
- Five classes – IgA, IgD, IgE, IgG, IgM
- Antibodies surround pathogens and neutralise them in various ways by binding to their antigenic sites





Learning Aim: Define "antibody" and discuss when antibodies are produced

Specific Immunity: How antibodies destroy pathogens

- Specific antibody is produced in response to the specific antigen for a specific pathogen after it invades.
  - EG: Measles antibody is produced in response to the antigenic site on the measles virus after it enters the body.
- Antibody binds to the antigenic site on the pathogen, neutralising it in a variety of ways:

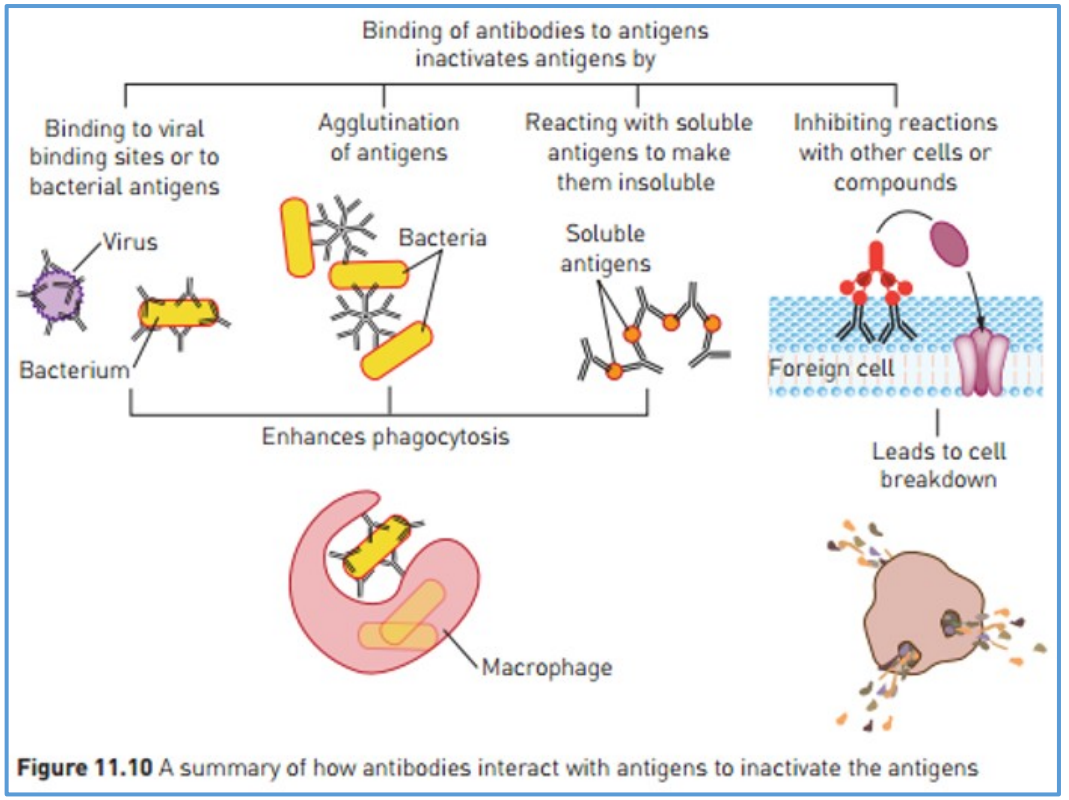


Figure 11.10 A summary of how antibodies interact with antigens to inactivate the antigens

1: Binding to active sites so virus/bacteria/toxin can no longer cause damage.

2: Sticking multiple pathogens together – *agglutination*

3: Reacting with soluble antigens to make them insoluble

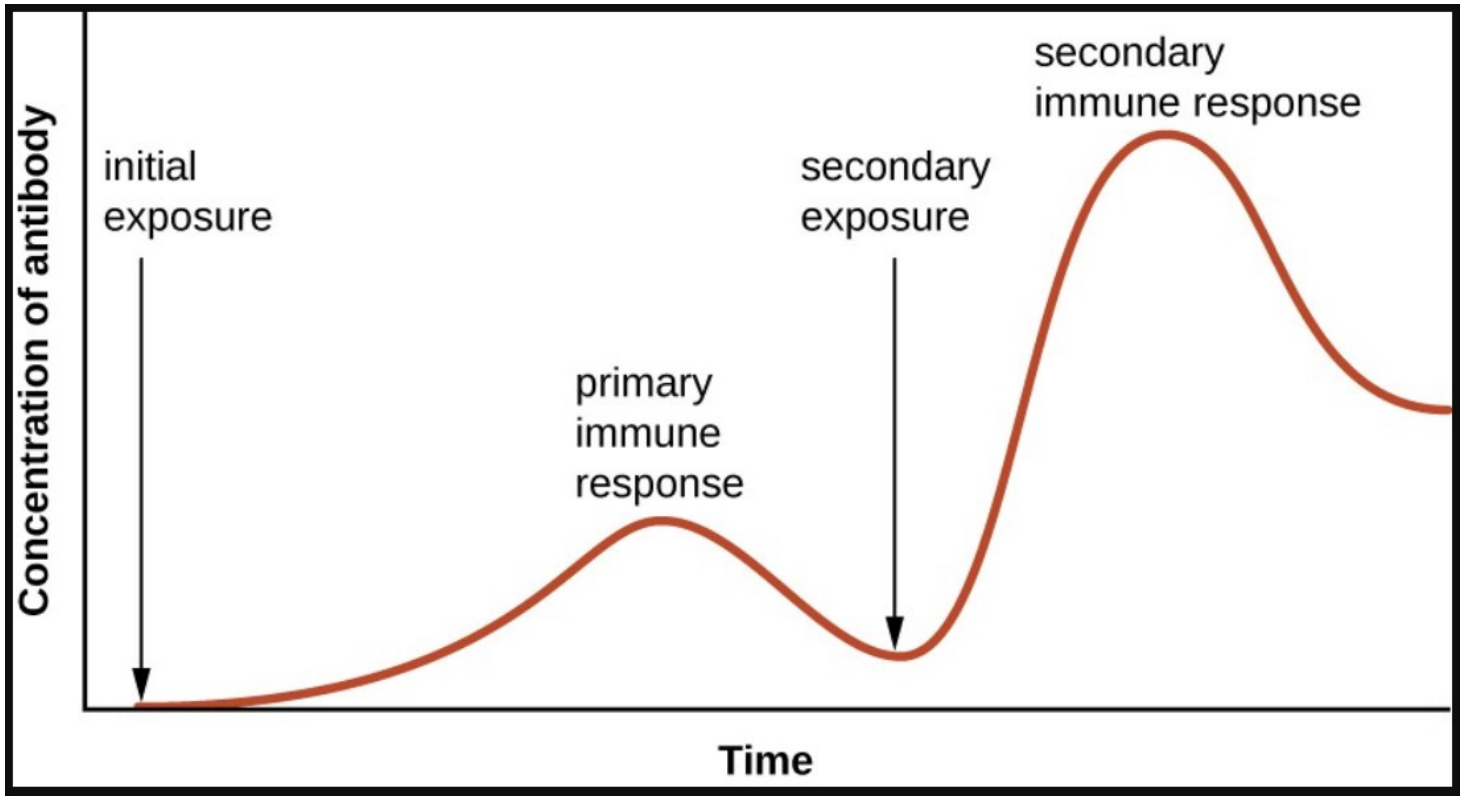
These enhance phagocytosis

4: Inhibiting reactions in foreign cells or compounds, leading to cell breakdown

Learning Aim: List and describe the ways in which antibody neutralises pathogens

Antibody concentration over multiple exposures to a specific pathogen

- *Primary Response*: the first time a specific antigen is encountered.
  - Slower response – takes several days to clone B-cells and manufacture antibody.
  - Symptoms occur and are then overcome.
- *Secondary Response:* subsequent times that the specific antigen is encountered.
  - Faster response – memory cells allow for faster recognition of antigen and therefore antibody release, before signs of illness develop.



Learning Aim: Describe differences between the first and second time the same pathogen is encountered.

Specific Immunity: Cell-Mediated Immunity

- Also known as T-Cell Immunity
- Involves T-Lymphocytes
  - Made in bone marrow
  - Mature in Thymus
- Deals with pathogens that have entered cells eg:
  - Viruses which replicate inside cells
  - Some bacteria that replicate inside cells (eg Tuberculosis)
- Deals with "abnormal" or "foreign" cells:
  - Fungi
  - Parasites
  - Tissue transplant rejection
  - Fighting cancer cells.

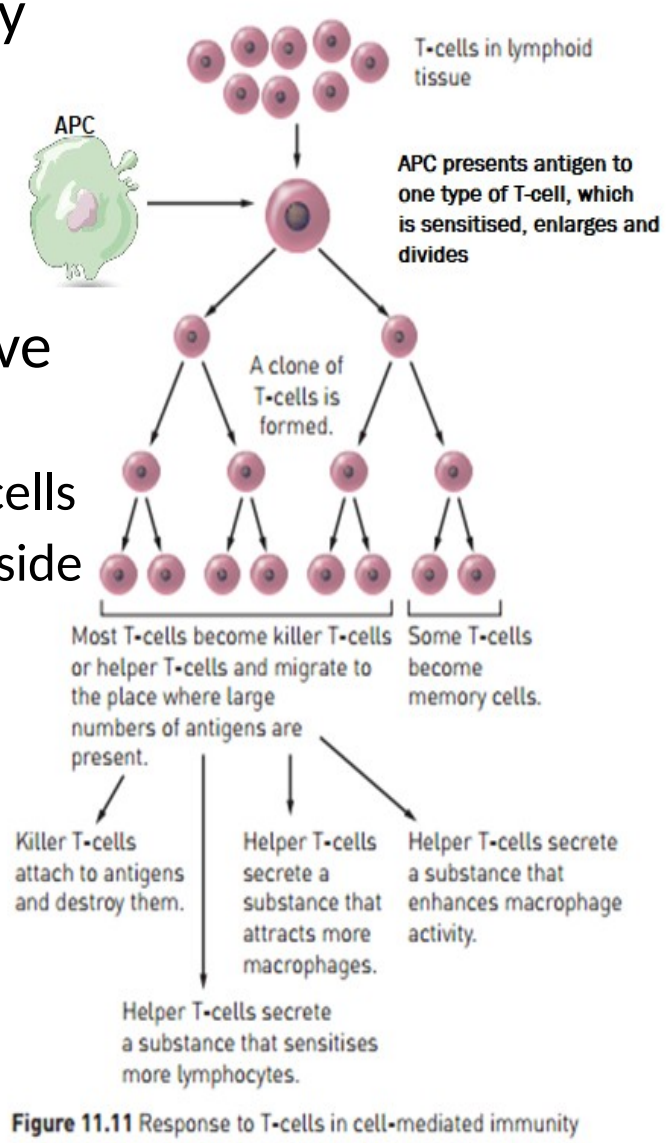


Figure 11.11 Response to T-cells in cell-mediated immunity

1: T-cells (T-lymphocytes) in lymphoid tissue.

2: APC has engulfed pathogen and presents antigen to one type of T-cell.

3: T-cell is sensitised, enlarges and divides.

4: Many clones of the T-cell are produced, sensitised for the antigen that was presented.

5: Sensitised T-cells can be:

- Killer T-Cells (Cytotoxic T-Cells)
  - Cells that have been infected have markers on their surface.
  - Killer T-Cells move to sites of infection
  - Attach to infected or foreign cells and destroy them.
- Suppressor T-Cells
  - Stop action of Killer T-Cells once infected cells have been destroyed.
- Helper T-Cells
  - Secrete substances that sensitise more lymphocytes, attract macrophages and enhance macrophage activity
- Memory T-Cells
  - Remain sensitised to the specific pathogen, in circulation and tissues, so that future responses to the pathogen are much more rapid.

Learning Aim: Describe in detail the function and processes involved in cell-mediated immunity

Table 11.1 Summary of immune responses

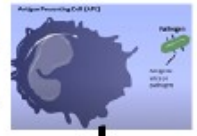
Antibody-mediated immunity (humoral immunity)	Cell-mediated immunity (cellular immunity)
Works against bacteria, toxins and viruses before they enter the body's cells; also against red blood cells of a different blood group than the person.	*Works against transplanted tissues and organs, cancer cells and cells that have been infected by viruses or bacteria; also provides resistance to fungi and parasites.*
1 Foreign antigen reaches lymphoid tissue. **2** Certain B-lymphocytes are stimulated to undergo rapid cell division. **3** Most new B-cells develop into plasma cells, which produce antibodies and release them into blood and lymph. **4** Antibodies combine with the specific antigen and inactivate or destroy it. **5** Some of the new B-cells form memory cells.	**1** Foreign antigen reaches lymphoid tissue. **2** Certain T-lymphocytes are stimulated to undergo rapid cell division. **3** Most new T-cells develop into killer T-cells or helper T-cells, which migrate to the site of the infection. **4** Killer T-cells destroy the antigen, while helper T-cells promote phagocytosis by macrophages. **5** Some sensitised T-cells form memory cells.

Learning Aim: Compare the processes of antibody-mediated and cell-mediated immunity

Specific Immunity

Simplified Flow Chart

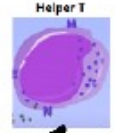
Antigen Presenting Cell (APC) encounters pathogen in lymph, engulfs it and presents antigen on surface.



T-lymphocyte binds to antigen presented by APC, and is sensitised, multiplies and differentiates into many different types of T-Cell



Helper T



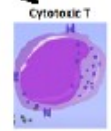
Helper T-Cells help to sensitise B cells

Suppressor T



Suppressor T-Cells stop Cytotoxic T – Cells from working once the infection is controlled.

Cytotoxic T



Cytotoxic T-Cells destroy infected body cells

B-Cell recognises the antigen due to unique receptors on its surface, is sensitised by a Helper T-Cell, and divides into Plasma cells and B-Memory Cells.

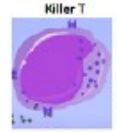


Plasma Cell

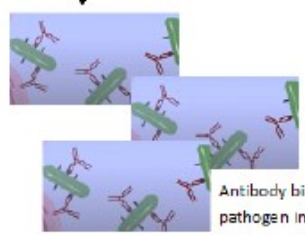


Plasma Cells produce antibody specific for the antigenic site of the pathogen encountered.

Killer T

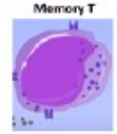


Killer T Cells assist in killing pathogens that have been neutralised by antibody



Antibody binds to pathogen in body fluids, neutralising it.

Memory T



Memory T-Cells retain the receptor for that antigen so that the response occurs faster if/when it is encountered again.

Memory B



Memory B-Cells retain the receptor for the specific antigen so that the response occurs faster if/when the pathogen is encountered again.

